COURSE

## IB Diploma Programme Business management HL

# Production and capacity

**Unit 5: Operations management · 5.2 Operations methods; 5.6 Production planning (HL only)**

Revision for Unit 5: Operations management: each firm works it out on its own deals.

Class **IB Business Management HL** Topic **Production and capacity**

THE LESSON

| | |
|---|---|
| **Setup** | 3 min |

**Round 1** — Decisions 3 min · Deals 5 min · Topic tasks 4 min

| | | | | |
|---|---|---|---|---|
| **Firm decisions** | **Deals** | TASK 1 **Capacity used %** | TASK 2 **Full capacity risks** | |

**Round 2** — Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 3 min

| | | | | |
|---|---|---|---|---|
| **Firm decisions** | **Deals** | TASK 3 **Fixed cost per car** | TASK 4 **New productivity** | **B2B deal** |

**Round 3** — Decisions 3 min · Deals 5 min · Topic tasks 4 min · B2B deal 3 min

| | | | | |
|---|---|---|---|---|
| **Firm decisions** | **Deals** | TASK 5 **Meet a short peak** | TASK 6 **Unit cost when full** | **B2B deal** |

| | |
|---|---|
| **Final questions** | 3 min |

Fits a 30-minute lesson setting; longer settings add more deals.

# Each step

| | |
|---|---|
| **Firm decisions** | Each role makes its decision on its own page. |
| **Deals** | Each pupil trades with a classmate in another firm. |
| **Topic task** | The firm answers together. Everyone presses Agree. |
| **B2B deal** | One firm sells a lot of cars to another firm. |
| **Final questions** | Each pupil answers six questions. |

# Firm decisions

| | | |
|---|---|---|
| **CEO** | **Staff** | Hire staff. Each costs a salary and gives one benefit. |
| | **Sabotage** FROM ROUND 2 | Hit a rival firm's cash, at a cost. A Legal Counsel blocks it. |
| **Sales & Marketing** | **Marketing** | Book one advert campaign to lift the next sales. |
| **Finance** | **Capital & Loans** | Borrow cash. The firm pays it back with interest. |
| | **Investments** FROM ROUND 2 | Buy property or funds. The value can rise or fall. |
| **Operations** | **Seasonal stock** | Order the season's cars in advance. |

**B2B deal · Rounds 2 and 3**

One firm sells a lot of 3 cars to another firm (4 in Round 3). With an odd number of firms, two firms offer to sell and the lower price wins.

# Topic tasks

Each firm gets its own figures. The method is the same for every firm.

## Task 1

ROUND 1

Workshop capacity.

**Capacity used, as a %?**

WORK OUT THE NUMBER · EXAMPLE FIGURES

| | |
|---|---|
| Prep bays | **3 bays** |
| Most one bay can do a week | **10 cars** |
| Cars prepared a week | **24 cars** |

METHOD

**Capacity used = cars prepared ÷ (bays × most per bay) × 100**

3 × 10 = 30; 24 ÷ 30 × 100 = 80%

**Why:** It has 6 cars of spare capacity a week.

## Task 2

ROUND 1

You run at 100% capacity all year.

**Which two problems are most likely?**

PICK TWO

- ○ A higher fixed cost per car
- ✔ **No spare time for repairs**
- ○ Spare capacity is wasted
- ✔ **Staff stress and more errors**
- ○ Staff with too little to do

**Why:** Full capacity spreads fixed costs well, but leaves no slack.

## Task 3

ROUND 2

Three workshops: fixed costs a year, and cars prepared.

**Order by fixed cost per car, lowest first?**

PUT THESE IN ORDER, FIRST TO LAST · RIGHT ORDER

1. **£45,000 fixed, 500 cars**
2. **£30,000 fixed, 300 cars**
3. **£20,000 fixed, 160 cars**

**Why:** £90, £100, £125 a car: the biggest costs, spread widest, are cheapest.

## Task 4

ROUND 2

More staff, more cars.

**New cars prepared per member of staff?**

WORK OUT THE NUMBER, THEN PICK IT · EXAMPLE FIGURES

| | |
|---|---|
| Cars prepared a week | **18 cars** |
| Staff | **3 staff** |
| Extra staff | **3 staff** |
| Extra cars prepared a week | **30 cars** |

METHOD

**Per staff = (cars + extra cars) ÷ (staff + extra staff)**

(18 + 30) ÷ (3 + 3) = 8 cars

**Why:** Productivity rose from 6 to 8 cars per worker.

## Task 5

ROUND 3

Demand is above your capacity for two months only.

**How should you meet it, and why?**

PICK A OR B AND GIVE A REASON

- ○ Build a new workshop
- ✔ **Subcontract the extra work**

**Why:** A short peak does not pay for new capacity.

## Task 6

ROUND 3

Run the workshop full.

**Fall in unit cost at full capacity?**

WORK OUT THE NUMBER · EXAMPLE FIGURES

| | |
|---|---|
| Fixed costs, a month | **£12,810** |
| Variable cost per car | **£400** |
| Cars prepared a month | **15 cars** |
| Most it can prepare a month | **35 cars** |

METHOD

**Fall = fixed costs ÷ vans now – fixed costs ÷ most vans**

£12,810 ÷ 15 = £854; ÷ 35 = £366; £854 − £366 = £488

**Why:** The variable cost stays; only the fixed cost per car falls.

# Final questions

**6**
questions for each pupil

**109**
Level 3 Business questions

**9**
of them on production and capacity

**The game draws 6 of the 109 Level 3 questions; these 9 are about production and capacity.**

1 A service centre can complete 640 jobs a month but completes 512. What is its capacity utilisation?
**80%.**

2 A parts team produces 1,440 tested units in 240 labour hours. What is labour productivity?
**6 units per labour hour.**

3 A bodyshop's practical monthly capacity is 900 repair-hours and booked work uses 765 hours. What is spare capacity as a percentage of practical capacity?
**15%.**

4 A workshop raises utilisation from 82% to 98%, while waiting time and rework rise sharply. What is the best interpretation?
**It is near its limit, so delays rise.**

5 Technician productivity differs widely between branches. Which study is most likely to support a fair operational decision?
**Compare job mix and hours first.**

6 Before a wellbeing redesign a workshop produced 450 standard hours with 90 paid hours; after, 520 with 104. What happened to measured labour productivity?
**It remained 5 standard hours per paid hour.**

7 A bodyshop can do 250 repairs a month and did 160. What is its capacity utilisation, and the effect on unit costs?
**64%: each repair carries more fixed cost than at full capacity.**

# Final questions

8 A fitting centre’s 27 fitters fit 918 tow bars a month. What is labour productivity?

**34 tow bars per fitter.**

9 A workshop can service 400 vehicles a month and serviced 300 last month: 75%. Which measure is that?

**Its capacity utilisation.**